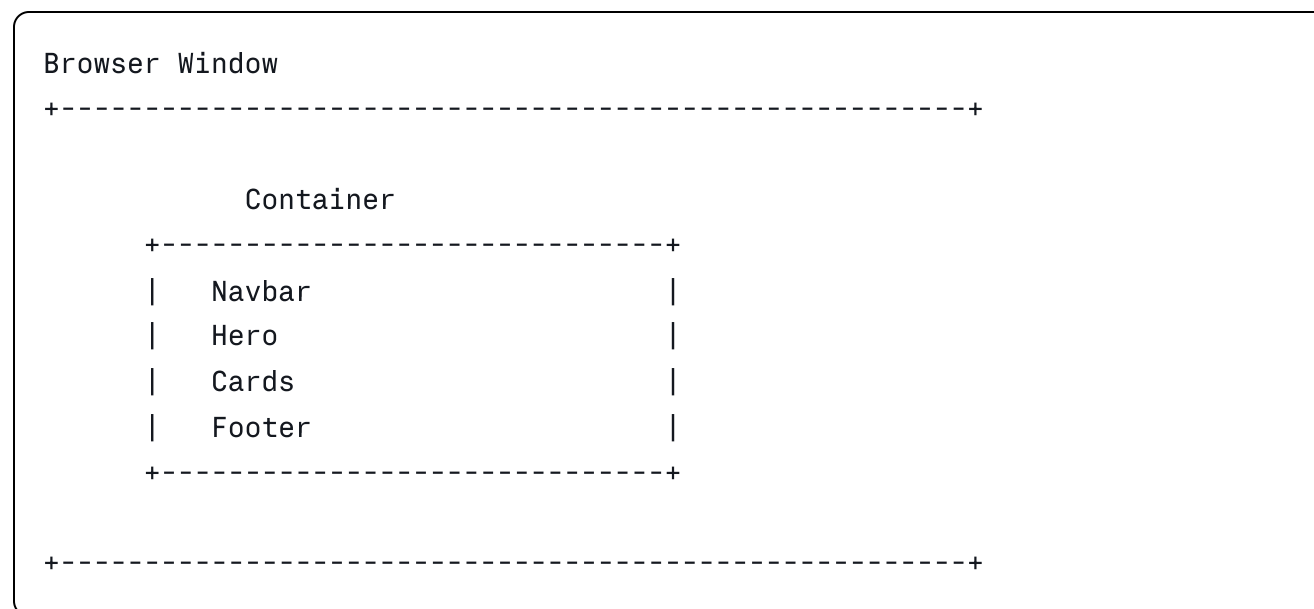


## Bootstrap — Topic 2: Container

Study paper format: notes (Roman Urdu + English technical terms) + active recall Q&A at the end.

## 1. Bootstrap Container kya hai?

**Container** ek wrapper hai jo website ke content ko hold karta hai aur proper spacing (padding) + width provide karta hai.



Bina container ke, content screen ke edges ko touch karta hai.

## Container kyun use karte hain?

- Content ko centered rakhne ke liye
- Left aur right padding add karne ke liye
- Maximum width control karne ke liye
- Layout ko responsive banane ke liye

## 2. Bootstrap Containers ki Types

Bootstrap **teen categories** provide karta hai:

1. `.container`
  2. `.container-fluid`
  3. Responsive Containers — `.container-sm`, `.container-md`, `.container-lg`,  
`.container-xl`, `.container-xxl`
- 

### 3. `.container`

Ye **sabse zyada use hone wala** container hai.

- Small screens par 100% width
- Larger screens par fixed maximum width
- Content hamesha centered rehta hai

```
html

<div class="container">
  <h1>Hello Bootstrap</h1>
  <p>This is inside a container.</p>
</div>
```

Large screen par dono sides pe empty space nazar aati hai — content center mein rehta hai.

### **.container** ki width (screen size ke hisab se)

| Screen Size | Container Width |
|-------------|-----------------|
| Extra Small | 100%            |
| Small       | 540px           |
| Medium      | 720px           |
| Large       | 960px           |
| Extra Large | 1140px          |
| XXL         | 1320px          |

Exact numbers yaad rakhna zaroori nahi — bas itna samjho ke screen barhne ke sath container bhi wider hota jata hai.

```
html

<div class="container bg-primary text-white p-5">
  This is a fixed-width container.
</div>
```

### **4. .container-fluid**

Ye container **hamesha browser ki full width** use karta hai — left/right koi empty space nahi hota.

```
html
```

```
<div class="container-fluid bg-success text-white p-5">
  Full Width Container
</div>
```

### Kab use karein:

- Hero sections
- Full-width banners
- Image galleries
- Google Maps
- Full-screen colored sections

```
html

<div class="container-fluid bg-dark text-white p-5">
  Hero Section
</div>
```

### **.container** vs **.container-fluid**

|                  |  |                         |  |                           |
|------------------|--|-------------------------|--|---------------------------|
| .container       |  | CONTENT                 |  | (centered, limited width) |
| .container-fluid |  | CONTENT CONTENT CONTENT |  | (full width)              |

## 5. Responsive Containers

Ye decide karne dete hain ke container **kis breakpoint se fixed-width** ban jaye.

Classes: **.container-sm**, **.container-md**, **.container-lg**, **.container-xl**, **.container-xxl**

| Class                       | Fluid rehta hai (breakpoint tak) | Uske baad |
|-----------------------------|----------------------------------|-----------|
| <code>.container-sm</code>  | <576px                           | Fixed     |
| <code>.container-md</code>  | <768px                           | Fixed     |
| <code>.container-lg</code>  | <992px                           | Fixed     |
| <code>.container-xl</code>  | <1200px                          | Fixed     |
| <code>.container-xxl</code> | <1400px                          | Fixed     |

```
html
<div class="container-lg">
  Large Container
</div>
```

Ye 992px tak fluid rahega, uske baad fixed width ho jayega.

## Breakpoints Reference

| Class | Breakpoint |
|-------|------------|
| sm    | ≥576px     |
| md    | ≥768px     |
| lg    | ≥992px     |
| xl    | ≥1200px    |
| xxl   | ≥1400px    |

Ye breakpoints Bootstrap mein har jagah responsive design ke liye use hote hain.

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## 6. Comparison Table

| Container Type                | Width Behavior                           |
|-------------------------------|------------------------------------------|
| <code>.container</code>       | Larger screens par fixed width, centered |
| <code>.container-fluid</code> | Hamesha 100% width                       |
| <code>.container-sm</code>    | 576px tak fluid, phir fixed              |
| <code>.container-md</code>    | 768px tak fluid, phir fixed              |
| <code>.container-lg</code>    | 992px tak fluid, phir fixed              |
| <code>.container-xl</code>    | 1200px tak fluid, phir fixed             |
| <code>.container-xxl</code>   | 1400px tak fluid, phir fixed             |

---

## 7. Complete Example

```
html
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>

  <div class="container bg-primary text-white p-4 mb-4">
    <h2>.container</h2>
    <p>Fixed width on larger screens.</p>
  </div>

  <div class="container-fluid bg-success text-white p-4 mb-4">
    <h2>.container-fluid</h2>
    <p>Always takes 100% width.</p>
  </div>

  <div class="container-lg bg-warning p-4">
    <h2>.container-lg</h2>
    <p>Fluid below 992px, fixed above 992px.</p>
  </div>

</body>
</html>
```

## 8. Konsa Container Kab Use Karein?

| Situation                | Best Choice                             |
|--------------------------|-----------------------------------------|
| Blog                     | <code>.container</code>                 |
| Portfolio                | <code>.container</code>                 |
| Dashboard                | <code>.container</code>                 |
| Landing page             | <code>.container</code>                 |
| Hero section             | <code>.container-fluid</code>           |
| Full-width banner        | <code>.container-fluid</code>           |
| Full-width image         | <code>.container-fluid</code>           |
| Custom breakpoint layout | <code>.container-sm/md/lg/xl/xxl</code> |

## 9. Interview Questions (as-is reference)

**Q1. Bootstrap container kya hai?** Ek responsive wrapper jo horizontal padding deta hai aur page content ki width control karta hai.

**Q2. `.container` aur `.container-fluid` mein farq?** `.container` = larger screens par fixed max-width, centered. `.container-fluid` = hamesha 100% viewport width.

**Q3. Responsive container kya hota hai?** Ek specified breakpoint tak full-width rehta hai, phir fixed-width container mein switch ho jata hai.

**Q4. `.container-fluid` kab use karna chahiye?** Un sections ke liye jo pura screen span karne chahiyein — hero banners, full-width images, maps, ya colored background sections.



## Practice Exercises

1. `.container` use karke ek page banao jismein heading, paragraph, aur button ho.
2. `.container-fluid` se ek full-width hero section banao — dark background, centered white text.
3. Hero ka `.container-fluid` replace kar ke `.container-lg` lagao aur browser resize kar ke **992px** breakpoint ke around width change observe karo.

Containers master karna zaroori hai kyunke almost har Bootstrap layout container se shuru hota hai, uske baad grid system (`.row` aur `.col-*`) uske andar aata hai.

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## Active Recall — Bootstrap Container

| Khud se answer karne ki koshish karo, phir neeche scroll karke check karo.

- Q1.** Container kya hota hai aur ye kya provide karta hai? Bina container ke content ka kya behavior hoga?
- Q2.** Bootstrap mein containers ki teen categories konsi hain?
- Q3.** `.container` ki behavior small screens par aur larger screens par kya hoti hai?
- Q4.** `.container-fluid` ki key characteristic kya hai, aur ye kin situations ke liye best hai?
- Q5.** `.container` aur `.container-fluid` mein core difference ek line mein likho.
- Q6.** Responsive containers (`.container-sm/md/lg/xl/xxl`) kaise kaam karte hain?
- Q7.** `.container-lg` ka breakpoint kya hai, aur uss breakpoint se pehle aur baad mein iski width kya hoti hai?
- Q8.** Sm, md, lg, xl, xxl breakpoints ki exact pixel values likho.
- Q9.** Blog aur Dashboard ke liye konsa container best choice hai, aur hero section ke liye konsa?
- Q10.** Bootstrap layout mein container ke baad typically kya aata hai?

---

## Answers

<details> <summary>Answers dekhne ke liye click karein</summary>

**A1.** Container ek wrapper hai jo content ko hold karta hai aur proper padding + width control provide karta hai. Bina container ke, content directly screen ke edges ko touch karta hai.

**A2.** `.container`, `.container-fluid`, aur Responsive Containers (`.container-sm`, `.container-md`, `.container-lg`, `.container-xl`, `.container-xxl`).

**A3.** Small screens par `.container` 100% width leta hai; larger screens par fixed maximum width ho jata hai aur content centered rehta hai.

**A4.** `.container-fluid` hamesha browser ki full width (100%) use karta hai, koi left/right empty space nahi hota. Ye hero sections, full-width banners, image galleries, maps, aur full-screen colored sections ke liye best hai.

**A5.** `.container` centered aur limited width deta hai, jabke `.container-fluid` hamesha full 100% width use karta hai.

**A6.** Ye apne specified breakpoint tak 100% (fluid) rehte hain, aur us breakpoint ke baad fixed-width ho jate hain.

**A7.** `.container-lg` ka breakpoint 992px hai. 992px se pehle ye fluid (100% width) rehta hai, aur 992px ke baad fixed width ho jata hai.

**A8.** sm  $\geq 576$ px, md  $\geq 768$ px, lg  $\geq 992$ px, xl  $\geq 1200$ px, xxl  $\geq 1400$ px.

**A9.** Blog aur Dashboard dono ke liye `.container` best hai. Hero section ke liye `.container-fluid` best hai.

**A10.** Container ke baad grid system aata hai — `.row` aur `.col-*` classes uske andar use hoti hain.

</details>